

XBOTIX – UNIVERSITY CATEGORY

Organized by:

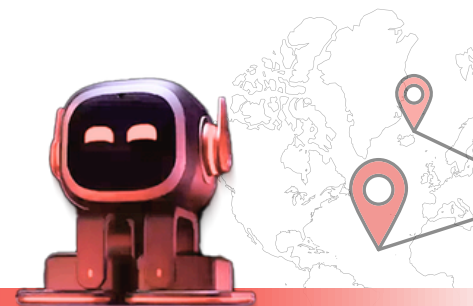
Electrical and Information Engineering Society In

collaboration with IEEE Student Branch, IET Ruhuna Chapter
and MMESS, Faculty of Engineering | University of Ruhuna



BACKSTORY

- Xbotix Galactic Core is the central AI nexus that **once powered peace, communication, and order across the galaxy.**
- It controlled the balance between intelligent machines and organic civilizations, maintaining harmony in every star system.
- Recently, a mysterious **corrupted signal** began spreading from deep space, affecting the Core's subsystems.
- As the Core loses control, robots across the galaxy malfunction, systems shut down, and planets start to break down.
- Entire colonies have gone dark, and the Core is nearing **critical failure**, the universe is close to a total system failure.
- In the ruins of the Xbotix Prime command station, one robot remains active, a small **guardian unit** called EMO.
- EMO was once part of the Core's defense system, but it survived the signal corruption due to an isolated subroutine.
- EMO finds a weak help signal from the center of the Core. It's a hidden message asking for help.
- EMO has limited power, with its limited power, EMO sends out a **beacon across the galaxy**, searching for skilled minds capable of restoring the Core.
- That beacon has reached **Earth... and you have received it**

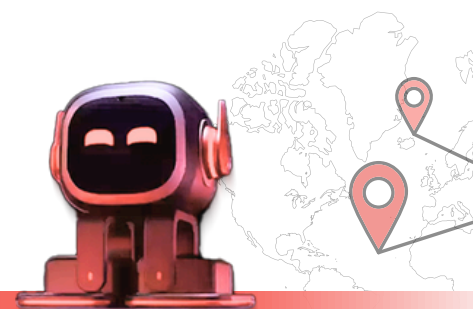


- The beacon carries more than a plea; it contains fragments of archived data, mission briefings, and tactical simulations from the **Xbotix Command Network**.
- Before humanity's brightest builders can take the next step, they must study this intelligence to understand the scale of the mission that lies ahead.

Mission Briefing – Xbotix Galactic Core Overview

Mission Simulation – Core Restoration Challenge

- Only by decoding these transmissions will the next generation of engineers be ready to answer the call when the galaxy needs them most.



SHARPEN - SCHOOL WORKSHOPS AND PRE-XBOTIX

- EMO's signal has reached Earth. Now, young minds are being awakened and trained to prepare for the restoration mission.

Stage Activities: Prepare, train, and activate the next generation of robotic minds by conducting school workshops and the pre-Xbotix.

SYNC - TEAM REGISTRATION, TESTING & QUALIFICATION

- With EMO's help, teams must now align with the Core protocols. Only the most stable and synchronized bots will be granted access to the Strike Phase.

Stage Activities: Team registration through the Xbotix website, team alignment, system checks, and mentoring.

STRIKE - XBOTIX FINAL COMPETITION & CORE ACCESS

- The corrupted Core lies ahead. Teams must now strike with precision, restore systems, and bring balance back to the galaxy.
- The best team will re-energize the Galactic Core and save the galaxy. Their skill, teamwork, and bravery will bring back order - and they will become legends across the stars.

Stage Activities: On the final competition day, the arena will be designed to show the steps of re-energizing the Galactic Core.

The final competition environment will be decorated according to the theme to represent the Galactic Core. A prototype of the EMO will be made to welcome the teams, express emotions, count down for the rounds, and show the leaderboard.





XBOTIX – UNIVERSITY CATEGORY



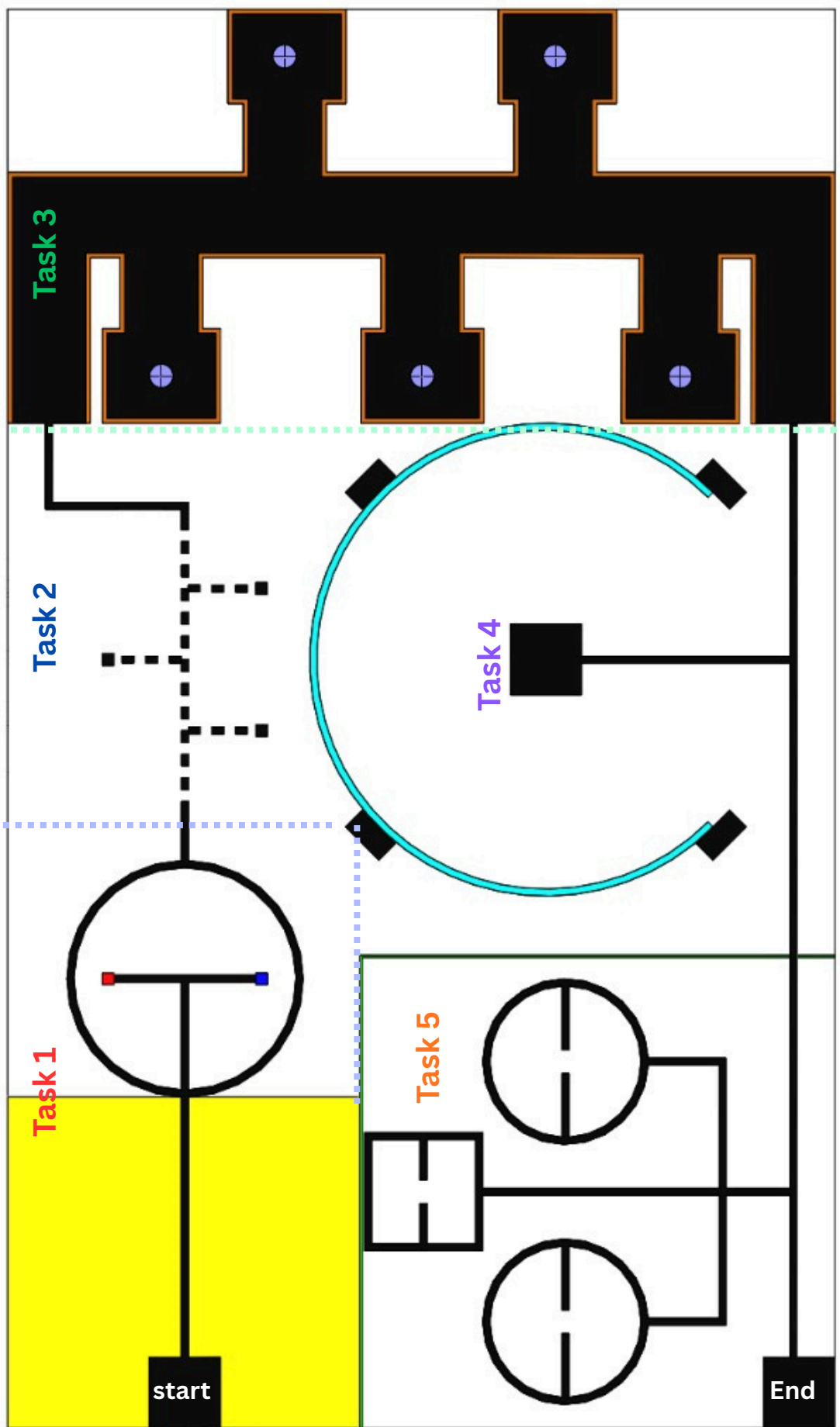
The art of robotics has leapt forward at light speed. Breakthroughs in AI, sensing, and autonomous control now allow young innovators attempt challenges once reserved for industry and space agencies. To ignite this passion in the next generation, the Xbotix team invites university builders to a mission unlike any other-**Xbotix: Galactic Core**.

Far beyond the Milky Way lies Xbotix Prime, a command station built around the universe's most powerful intelligence: the Galactic Core.

Now, that intelligence is under threat. A mysterious corrupted signal has swept in from deep space, causing subsystems to fail and civilizations to vanish into darkness. As chaos spreads, the Core's defenses have transformed into a living firewall of **five sealed sectors** that protect its fading heart. If the Core falls silent, every star-linked world will spiral into chaos.

Only one sentinel remains awake, EMO, a small guardian robot whose isolated subroutine shielded it from the corruption. EMO cannot restore the Core alone. Its final encrypted beacon has crossed galaxies and reached Earth, summoning the **brightest young builders** to join the mission.





Task Description



The Xbotix 2025 consists of five key tasks which, the robot Ares must autonomously complete:

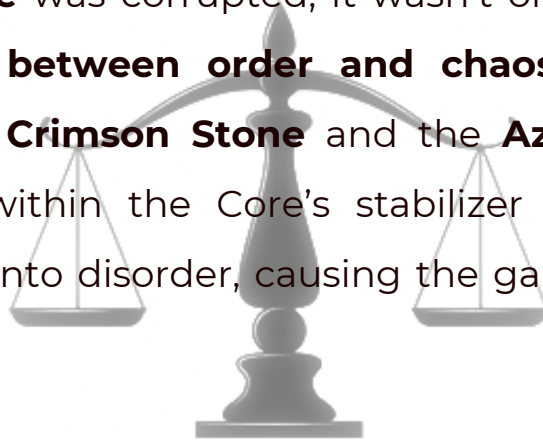
1. **Task 1: The Stabilizer Trials** - Cube collection and placement zone
2. **Task 2: Quantum Towers Protocol** - Three luminous towers arranged in randomized positions.
3. **Task 3: Firewall of Harmony** - Wall-following path with five towers presenting energy cells.
4. **Task 4: Firewall Storage Protocol** - Circular throwing area with four labeled storage boxes (A–D).
5. **Task 5: The Last Energy Cell** - Placement zone for the last energy cell.

ROUND 01 - ELIMINATION



Task 1-The Stabilizer Trials

When the **Galactic Core** was corrupted, it wasn't only its power that failed — the **balance between order and chaos** collapsed. Two ancient keystones, the **Crimson Stone** and the **Azure Stone**, once maintained harmony within the Core's stabilizer system. Without them, the Core drifted into disorder, causing the galaxy's equilibrium to falter.



To restore balance, the robot must embark on the **Path of Equilibrium**. A journey through unpredictable terrain designed to test precision, decision-making, and stability.





The robot begins by following a **line-guided path** whose shape is **unknown**. Along this path lie **curves, dead ends, and ramps** (Appendix 2), remnants of the Core's fractured geometry. Somewhere within this undefined route, the robot will locate the two lost keystones: (Appendix 1)

- The Crimson Stone (**red cube**) – symbol of chaos.
- The Azure Stone (**blue cube**) – symbol of order.

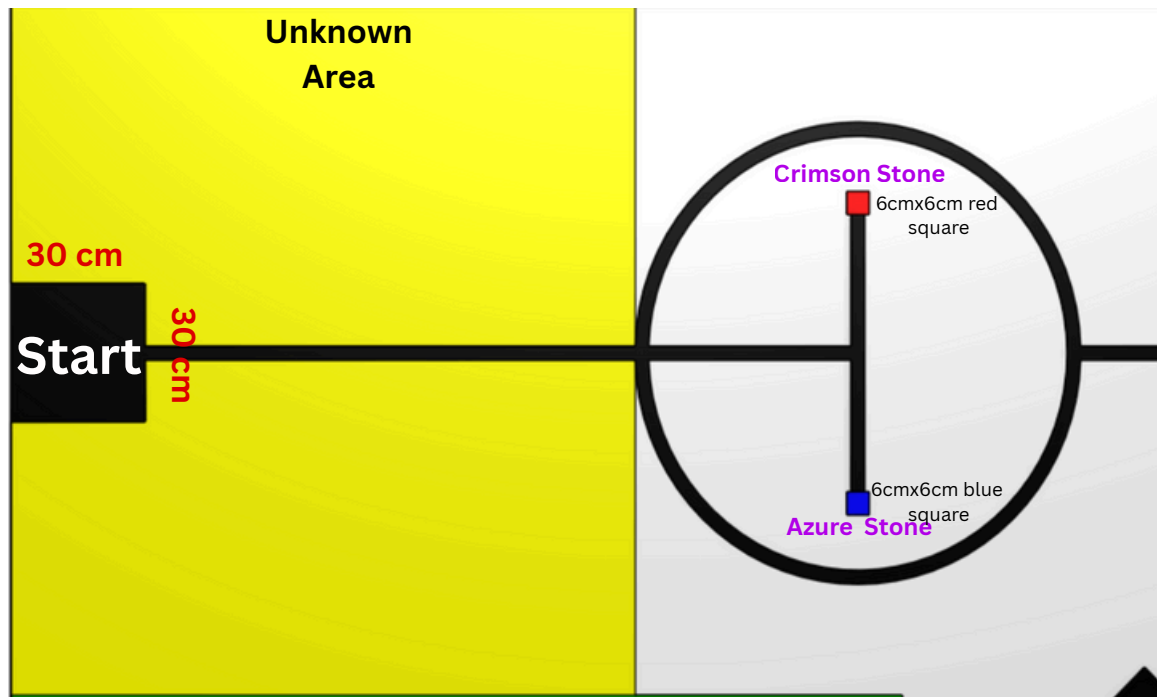
After both cubes are recovered, the robot proceeds to the **Circular Chamber (with 100cm diameter)** at the end of the path. At the center, the route divides into two separate branches, each leading to a Stabilizer Socket:

- The Crimson Stone (red cube) - **left-side** stabilizer socket.
- The Azure Stone (blue cube) - **right-side** stabilizer socket.

When both cubes are correctly positioned, the Core's harmonic field will begin to realign, partially restoring the system's balance. Once this equilibrium is achieved, the robot must return to the main path and continue toward the next mission in the Galactic Core restoration sequence.

The cube **unloading areas**(stabilizer sockets) would be **6cmx6cm** squares. For Crimson stone the socket would be in **red color** and the Azure stone the socket would be in **blue color**.





****Based on the performance in the elimination round, only the top 15 teams will qualify for Round 2, where they will proceed to attempt all designated tasks.**

ROUND 02

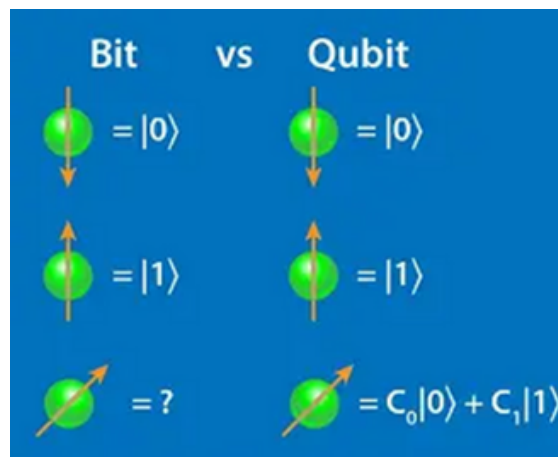


Task 1-The Stabilizer Trials

For the second round, Task 1 – The Stabilizer Trials will remain identical to the elimination round. This allows the selected teams to further refine their performance, improve consistency, and enhance stability. In Round 2, teams will proceed through all designated tasks, building upon the foundation established during the elimination phase.

Task 2 (Quantum Towers Protocol)

Now, the robot enters the Quantum Computing Section. You have to get some data from the quantum computer to recover the system. It will be a 3-bit binary value. Within the arena exist **three luminous towers**, each symbolizing a quantum bit (qubit).



In conventional binary logic, a bit is deterministically 0 or 1. By contrast, qubits possess additional structure: they may exist in basis states $|0\rangle$ (mapped here to **red emission**) or $|1\rangle$ (mapped here to **green emission**), but prior to interrogation they may also inhabit a **superposed configuration**, visually represented by the absence of illumination.

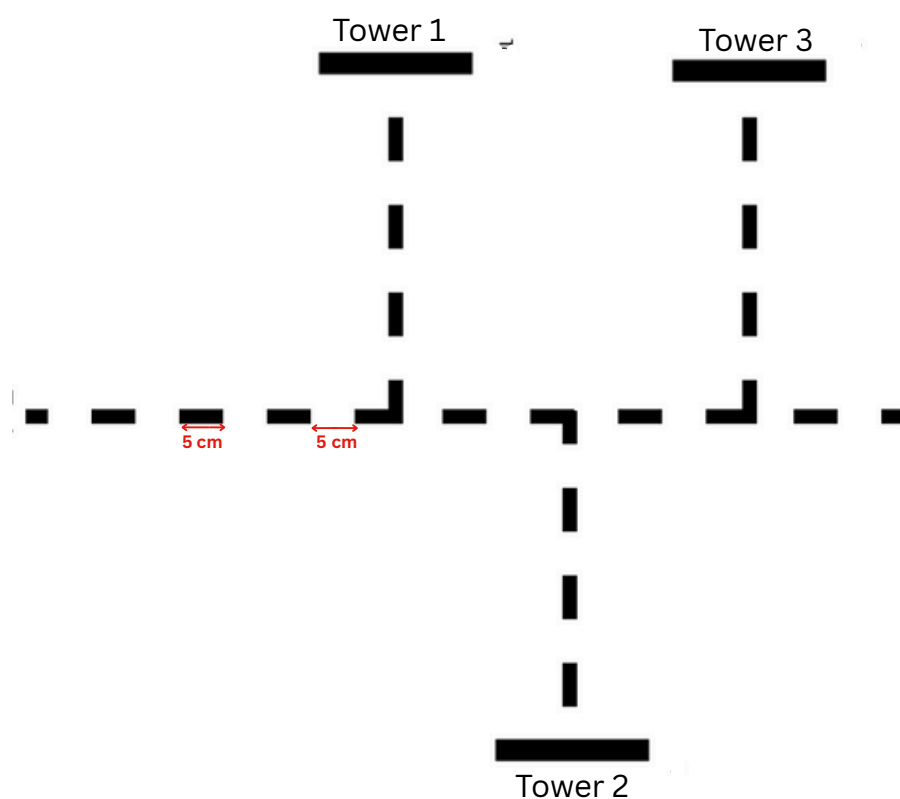
Measurement transforms this indefinite superposition into a definitive eigenstate, irreversibly fixing the qubit into $|0\rangle$ or $|1\rangle$. In this challenge, **illumination corresponds to classical outcomes**, while darkness corresponds to the unresolved quantum state.

- **Two towers** are initialized in determinate eigenstates ($|0\rangle$ or $|1\rangle$), which are **visually available** at the commencement of the task. These require no intervention.
- **One tower** begins in the unresolved state (lamp OFF). This condition implies superposition. A **manual perturbation** (button actuation) triggers the measurement process, forcing state reduction into either red (0) or green (1). (**Appendix 3**).

Once this collapse occurs, all three towers collectively encode a **3-bit register**.

Operational Directive

1. The robot must acquire the **visible binary assignments** of the two already-measured qubits.
2. For the third (unmeasured) qubit, the robot must actuate the control interface (button).
3. Upon actuation, the qubit's superposed state collapses into one of its computational basis states, after which the robot must register its observed value.
4. The resultant triplet of values (from the two fixed qubits and the one newly collapsed qubit) constitutes a **quantum-derived binary string** to be stored in memory.



Example,

- Tower A → luminous green → interpreted as $|1\rangle$.
- Tower B → luminous red → interpreted as $|0\rangle$.
- Tower C → unlit → post-actuation collapse → luminous red → interpreted as $|0\rangle$.

Thus, the composite register = 100.

The robot must rely exclusively on its own perception and internal memory to construct the binary string; no external correction or re-measurement will be available.

Non Illuminated lamp is connected with a push button. The touching area of the push button would be 3 cm diameter. The push button in the switch is indicated by the circle which is 5.0 cm above the platform's surface. **(Appendix 4)**

Colours will be randomized, so **sometimes two teams may not encounter the same color arrangement.**

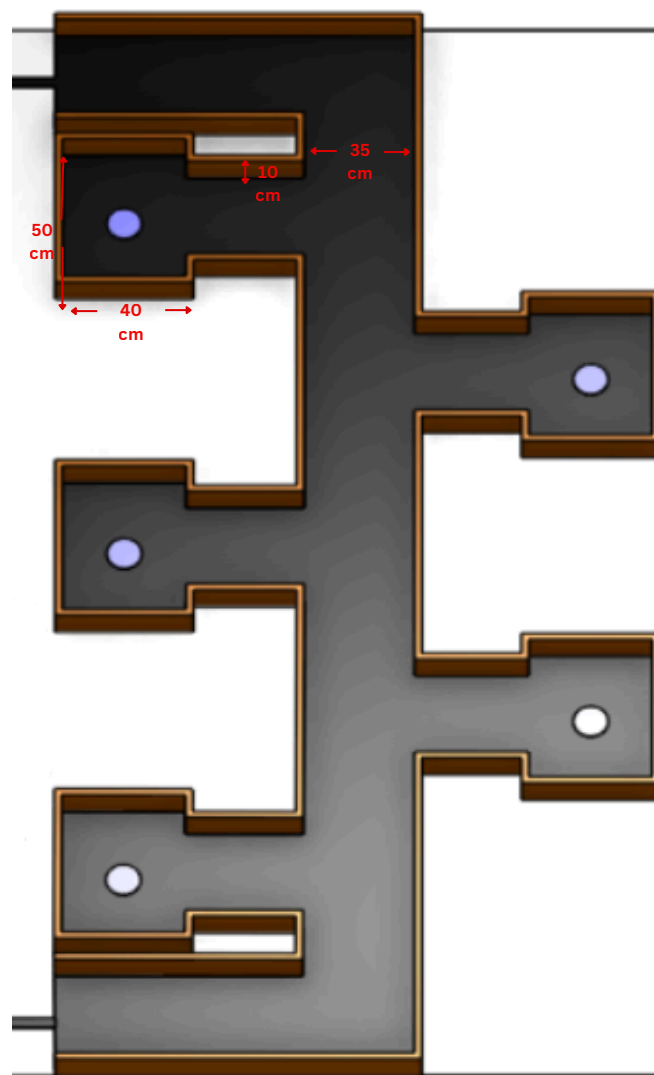
The connecting paths will be shown with dashed lines. Every turn in the arena follows an **'L' shape, with no 'T' junctions present**

There should be a clear method of indication of the detected color from towers. the method of indication should be declared to the judges before starting the attempt.

Task 3: Firewall of Harmony

Beyond the Four-Way Nexus lies the **Firewall Zone**, a defensive corridor built long ago to protect the Galactic Core from corrupted intrusions. Over time, the firewall itself has fractured. Its towers now scatter fragments of the Core's ancient harmony into glowing spheres known as **Harmonic Fragments**.

- **Green fragments** resonate with balance and order. (Appendix 5)
- **Red fragments** carry the corrupted echoes of chaos.
- To restore the Core's stability, the robot must collect these fragments and decode their meaning. The number of green fragments holds the **Harmony Code**. A binary sequence that can be fed back into the Core's restoration protocols. The robot enters the **Firewall Zone** and follows the wall path.



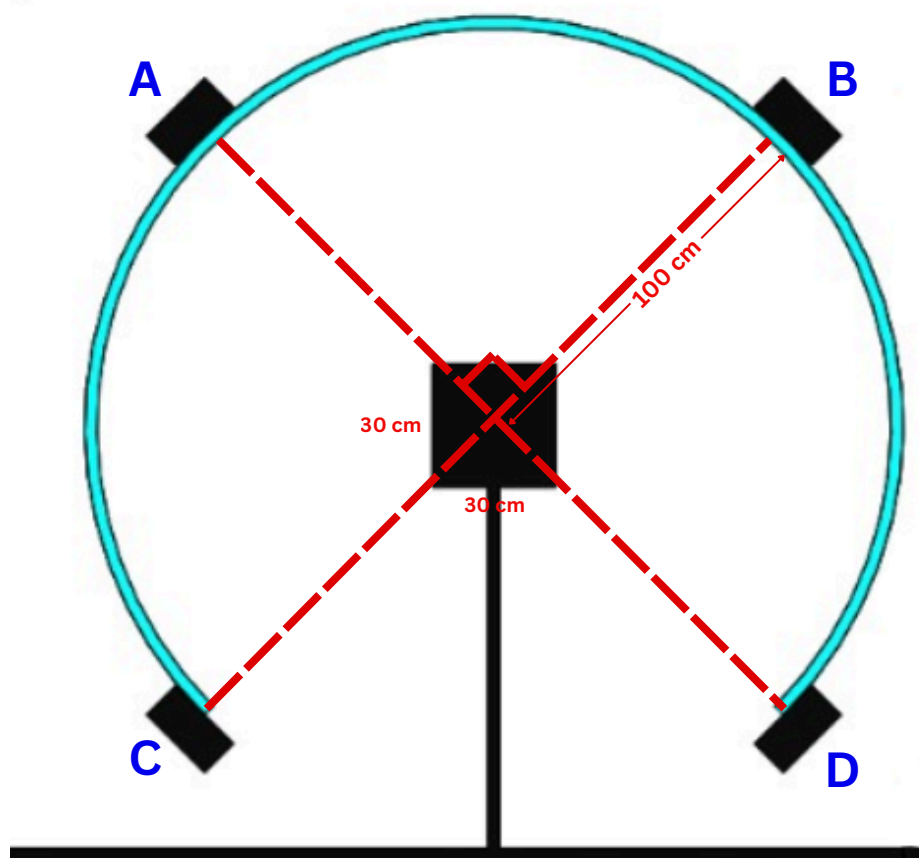
- The robot must collect **all five fragments**. If the robot fails to grip the ball or drops it, it must **restart from the starting point of Task 3**.
- Along the path, five towers will present glowing **Harmonic Fragments** (spheres).

- As each is collected, the robot must:
 - Identify its color (green or red).
 - Keep track of how many **green fragments** are found.
- After collecting all fragments, the robot must:
 - Count the total green fragments.
 - Convert this count into its **binary equivalent**.
 - Store this binary value in its internal memory.

Note: Red fragments represent corrupted signals and **do not contribute** to the Harmony Code.

The final binary value represents how much balance was recovered from the Firewall Zone.

Task 4 : Firewall Storage Protocol



In this section, the robot is required to store the **Harmonic Fragments**, which are represented by the balls collected during Task 3. Unlike previous tasks, the robot cannot physically reach the storage boxes. Instead, the robot must transport the collected fragments to a designated throwing position located at the center of the circle. From this position, the robot must identify the storage boxes and accurately throw each fragment into the corresponding box(color of the box will be gray in color). Successful completion of this task requires the robot to correctly recognize the target boxes and ensure that all collected fragments are deposited inside them.

In this task, the robot must store the collected fragments, which consist of two types: green (order) and red (chaos). The placement of these fragments into the storage boxes follows a specific computational process. From Task 2 and Task 3, the robot has already obtained two sets of three-bit Harmony Codes. These values shall be arranged in the form of a 2×3 matrix. This matrix must then be multiplied by the **Transpose Matrix** of the arranged 2×3 matrix, resulting in a 2×2 matrix. The robot must then calculate the summation of all four elements of this 2×2 matrix. The resulting number is divided by 4, producing two results: the **quotient** and the **remainder**.

The storage rule is as follows:

- **Green fragments (order)** must be thrown into the storage box corresponding to the **quotient**.
- **Red fragments (chaos)** must be thrown into the storage box corresponding to the **remainder**.

Thus, the robot must use its computation, perception, and throwing mechanism to correctly identify the storage boxes and deposit the fragments in the correct order. (**Appendix 6**)

A	0
B	1
C	2
D	3

Example :

$$\begin{array}{l} \text{Task 2} \rightarrow \\ \text{Task 3} \rightarrow \end{array} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$Y = (1 + 0 + 0 + 2) / 4$$

$$Y = 3 / 4$$

$$\text{Reminder} = 3$$

$$\text{Quotient} = 0$$

$$\text{Green Fragment} = A$$

$$\text{Red Fragment} = D$$

Task 5 : The Last Energy Cell

Deep within the ruins of the Galactic Core, silence echoes through the broken halls. The great engines that once pulsed with cosmic energy now lie dormant. Entire star systems wait in darkness, their survival depending on a single spark of power.

Legends speak of two ancient energy cells hidden within the chamber. One still brims with life, glowing faintly with the remnants of stored starlight. The other, however, has long since faded an empty husk, drained by time and corruption. Only the **true, charged cell** can awaken the Core once more. (Appendix 7)

But the challenge is not as simple as retrieval. The cells are scattered across different paths, and only one carries the vital charge. Your robot, acting as the lone guardian, must **search the arena and identify the living cell. Weight of the both cells will be around 80g and the surface of both cells will be Blue.** The discovery must be clear and decisive signaled through a buzzer, a flashing light, or another unmistakable mark of recognition.

Even then, the trial is not complete. The Core's **cell slot demands precision.** Its polarity is predetermined, and the cell must be aligned perfectly.

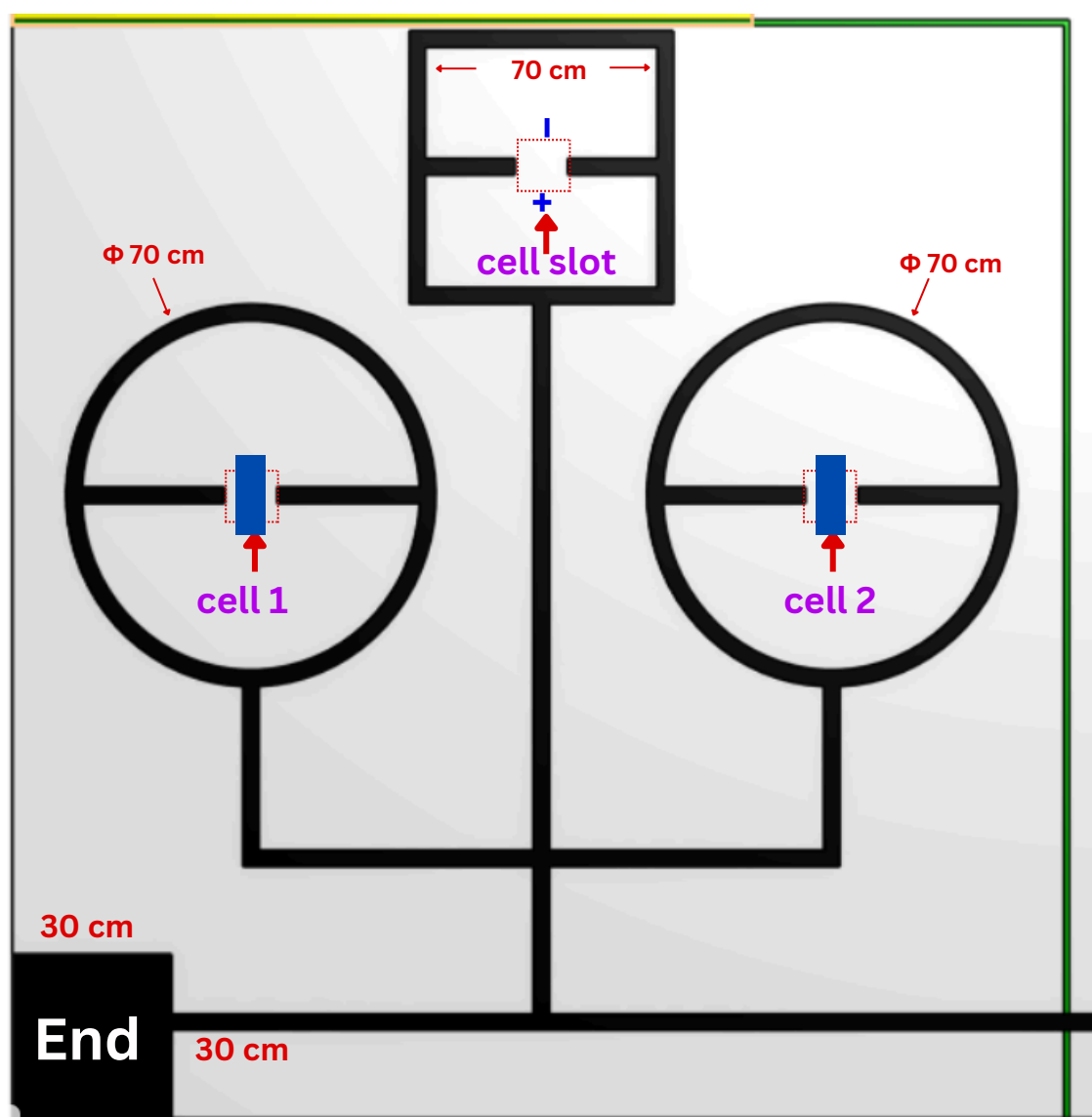
A wrong placement would mean failure; the Core would remain silent, and the galaxy would sink deeper into shadow.



Once the correct cell is carried across the arena and carefully placed into the slot, the chamber will respond. Lights will surge back to life, conduits will blaze with energy, and the **Galactic Core will stir from its slumber**. The robot's courage and accuracy will restore balance to the stars.

This is not merely a task.

It is the **moment of rebirth** where your robot's choices decide the fate of the galaxy itself.



Arena Specifications



- The **Start** and **End** squares shall be **black squares of 30 cm × 30 cm**.
- All **line segments** shall be **black lines with a width of 3 cm** on a white background, unless otherwise specified.
- The arena shall be constructed using multiple boards; therefore, **slight differences in height** may occur at the boundaries of the boards.
- The **minimum distance** between the edges of any line and the boundary walls of the arena shall be **15 cm**.
- The surfaces of all arena components, including **lines, walls, and ramps**, shall be **matte-finished** to prevent glare and sensor interference.
- The total arena dimensions shall be **600cm x 350cm**.
- **Dashed line guides** will be used in the Quantum Towers section, and walls will be present in the Firewall Zone.
- **Dead ends, ramps, and curved paths** may be introduced to test navigation.
- All dimensions mentioned throughout this document are subject to a maximum error margin of **±0.5 cm**.



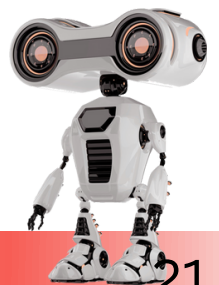
Build Requirements

Teams must design and construct their robots from scratch. Off-the-shelf robotic kits or complete pre-assembled platforms are not permitted. However, teams may use standard electronic components such as microcontrollers, sensor modules, motors, wheels, gears, and gripper mechanisms, provided they are integrated into a custom design suited for the competition tasks, including cube handling, button actuation, color detection, wall following, ball throwing, and final energy cell placement.

Teams must implement a clear visual indication system on their robot to display the recognized 3-bit binary value obtained from the quantum towers. The indication method should be declared to the judges before the attempt start.

Physical Constraints

The robot's base footprint must not exceed **25 × 25 cm** at the start of the match. Height is unrestricted, and mechanisms such as arms, grippers, or throwing devices may extend during operation, but must retract to fit the initial size limit before activation. Robots must remain a single integrated unit and are not permitted to detach, separate, or leave any components behind in the arena during a run.



Power & Voltage

Robots must operate using onboard batteries or rechargeable cells only. The maximum allowable system voltage is 24 V. All power sources must fit within the robot's defined constraints. Each robot must include a single, easily accessible power switch for activation.

Autonomous Operation

Once activated, the robot must function entirely on its own, without any form of external control. Any use of wired or wireless communication or manual intervention will result in immediate disqualification.

Arena Compatibility

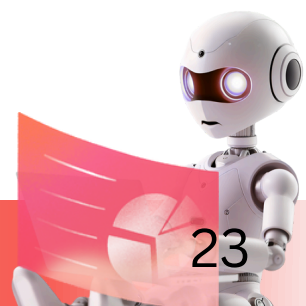
Design your robot to avoid damaging the competition surface. Scratching, marking, or leaving debris will result in disqualification. Officials may reject robots they deem unsafe for the arena. Robots must stay intact throughout the match; no dropping parts, separating into multiple units, or leaving components behind. Build with sufficient edge detection or balance to prevent falling off the platform. Ensure your design works under normal indoor lighting conditions without requiring special environmental setup.



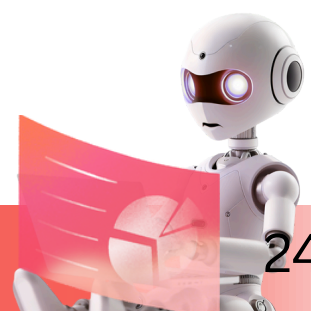
Rules and Regulations



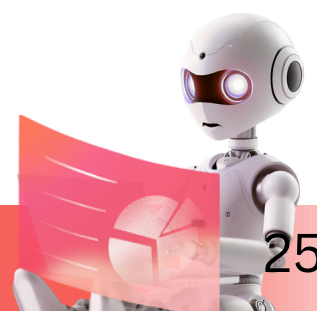
- All the robots are collected 15 minutes before the competition.
- Once a round has begun, no changes will be made to the arena layout. Every team will compete in the exact same arena design to ensure fairness.
- Restarting the robot from a checkpoint will be considered as starting a new attempt, and it will count toward the total number of attempts allowed.
- Teams must be ready to begin their run within five minutes of being called. If they are not ready within this time, their attempt will be considered lost.
- At the start of each run, the robot must be placed completely within the starting square. After the judges give the signal, the robot may be switched on, and from that point onward, it must move and operate on its own. Contestants are not allowed to physically adjust the robot's orientation or communicate with it in any way during the attempt.
- Each team is allowed three attempts within a single round. The best attempt will be considered for grading.



- Maximum time of **8 minutes** is allocated for a team in Round 1 **including the 2 minutes for calibration.**
- Maximum time of **15 minutes** is allocated for a team in Round 2 (for all three attempts, **excluding 2 minutes calibration time**).
- Modification of the the hardware(battery) of the robot is allowed only between the interval of the first round and the second round.
- If the robot exceeds the 15-minute time limit, it will be removed from the arena.
- No programming or hardware changes are allowed during the 15-minute gameplay window. Once the round begins, the robot must perform with its existing setup.
- The total run time is measured as the time taken for the robot to travel from the Start square to the End square for Round 2.
- The timer begins when the judges signal the start and stops only when the robot completes its run. The timer will not be paused for any reason during the run.



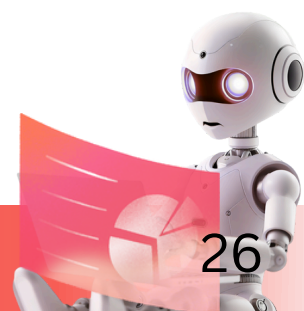
- If the robot leaves the line completely so that no part of it is touching the line, the judges will count it as going off-track.
- If the robot skips more than 30 cm without returning to the line, the attempt will be ended, and the robot will be removed from the arena.



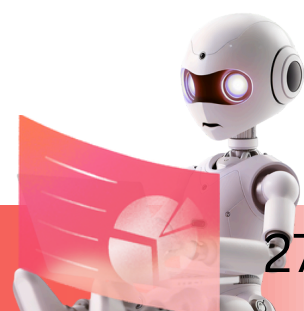
Calibration Rules



- Each team will be given an additional two minutes before the 15-minute gameplay period to calibrate their robot.
- Only external mechanical adjustments to the robot are allowed during calibration. Teams are not permitted to make programming changes or replace any hardware components during this time.
- If a team is ready, they may request to start their first attempt before the two-minute calibration period has ended. If they do this, the 15-minute gameplay period will begin immediately, and any unused calibration time will not be added later.
- If a team takes longer than the given two minutes for calibration, the extra time used will be deducted from their 15 minutes of gameplay. This ensures that no team receives more time than others.
- A team may stop the robot at any point and be credited with the distance traveled and the time consumed until the stop.



- The organizing committee and officials are not responsible for any damage to robots caused during calibration or competition.
- Any questions regarding the tasks or rules must be raised formally by the team leader to the organizing committee, and the judges' decision will be final and binding.

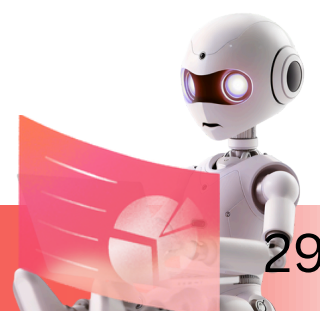


Participation Rules & Team Setup



- Each team must consist of a maximum of **5 members**, including the team leader.
- The team leader will be the official representative of the team and the only person allowed to raise objections, clarifications, or appeals to the organizing committee.
- Every participant must be officially registered under their respective institution and provide valid identification upon request.
- Teams must report to the competition area at least **30 minutes before their scheduled slot**. Any team failing to report on time may be disqualified or have their attempt forfeited.
- A maximum of **one robot per team** is permitted for the competition. Backup robots are not allowed; however, spare parts and replacement components may be used between rounds.
- Teams are responsible for transporting, maintaining, and operating their robots throughout the event. The organizers will not provide additional tools, components, or technical assistance.

- Cooperation between teams for the purpose of gaining an unfair advantage is strictly prohibited and may result in disqualification of all parties involved.
- All team members must be present during the competition. Substitution of participants is not allowed unless formally approved by the organizers in advance.
- The safety of participants, judges, and the audience must always be maintained. Any robot deemed unsafe by the officials will not be permitted to compete.
- Teams must respect the competition schedule and instructions from organizers. Failure to comply may result in penalties, disqualification, or loss of an attempt.

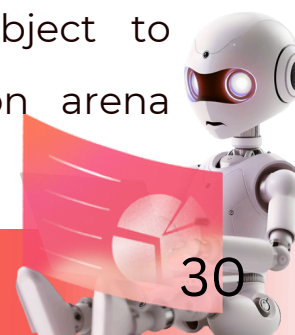


Scoring & Assessment



Task	Points
1 – The Stabilizer Trials	150
2 – Quantum Towers Protocol	150
3 – Firewall of Harmony	200
4 – Firewall Storage Protocol	250
5 – The Last Energy Cell	250
Time Bonus	100

- Marks will be allocated based on successful completion of each task as specified.
- Additional marks may be influenced by judges' evaluation of navigation quality, sensing accuracy, computational correctness, and mechanism precision.
- A time bonus of up to 100 points will be awarded only if all five tasks are successfully completed.
- Penalties may be imposed for arena damage, unsafe operation, or any behavior that risks fairness. Severe cases may lead to immediate disqualification.
- Any form of human intervention, remote control usage, or violation of autonomous operation rules will result in disqualification.
- Please note that this scoring scheme may be subject to modification during the final competition based on arena conditions, and the judges' decisions will be final.



- All robots will be evaluated on their ability to complete the five designated tasks autonomously within the given arena. Evaluation will consider not only task completion but also accuracy, stability, and compliance with the specified rules. Judges will closely observe task execution to ensure that actions such as cube placement, button actuation, color detection, wall-following, ball throwing, and energy cell handling are performed strictly by the robot without external assistance.
- Robots will be assessed on **technical performance** (precision of movements, reliability of sensors, and successful execution of computations), **arena compatibility** (safe interaction with surfaces, ramps, and walls), and **design compliance** (size, weight, and power restrictions). Any robot that causes deliberate damage to the arena, bypasses tasks, or violates physical or operational constraints will be penalized or disqualified.
- Scoring will be strictly based on the official marking scheme, with deductions applied for errors, collisions, missed objectives, or violations. Partial points may be awarded for incomplete task execution if the robot demonstrates progress toward the objective. Judges will also verify that recorded binary values, matrix calculations, and final placements are correct as per the rules. All evaluations will be carried out by the competition officials, and their decisions are final and binding.



Special note



- Teams are advised not to make assumptions about the arena or the tasks beyond what is described in this document. For any uncertainties or clarifications, please reach out to the organizing committee directly. This is the second release (Version 2) of the task document. Further announcements or revisions will be communicated through the official WhatsApp group and the Xbotix website.

Contact Information



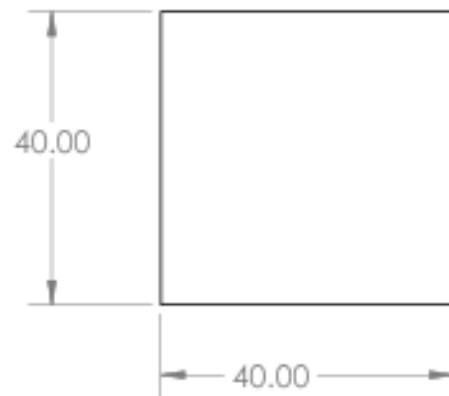
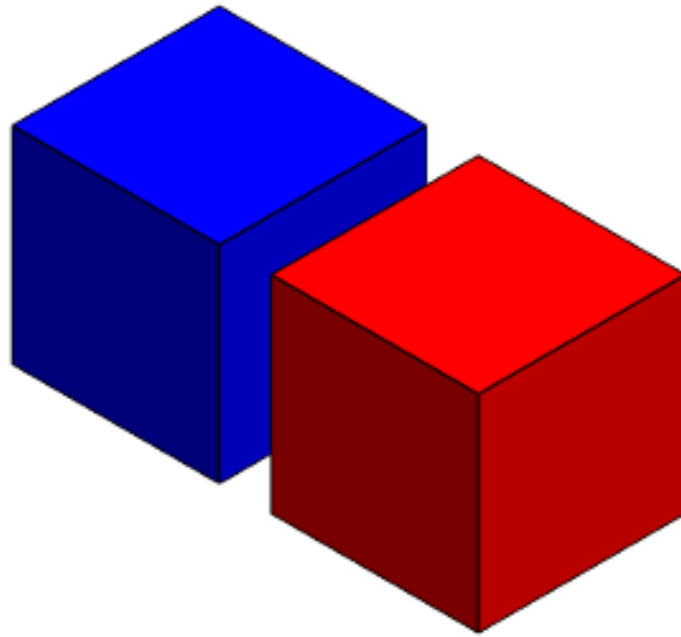
 help.xbotix@gmail.com

 +94 76 579 8787

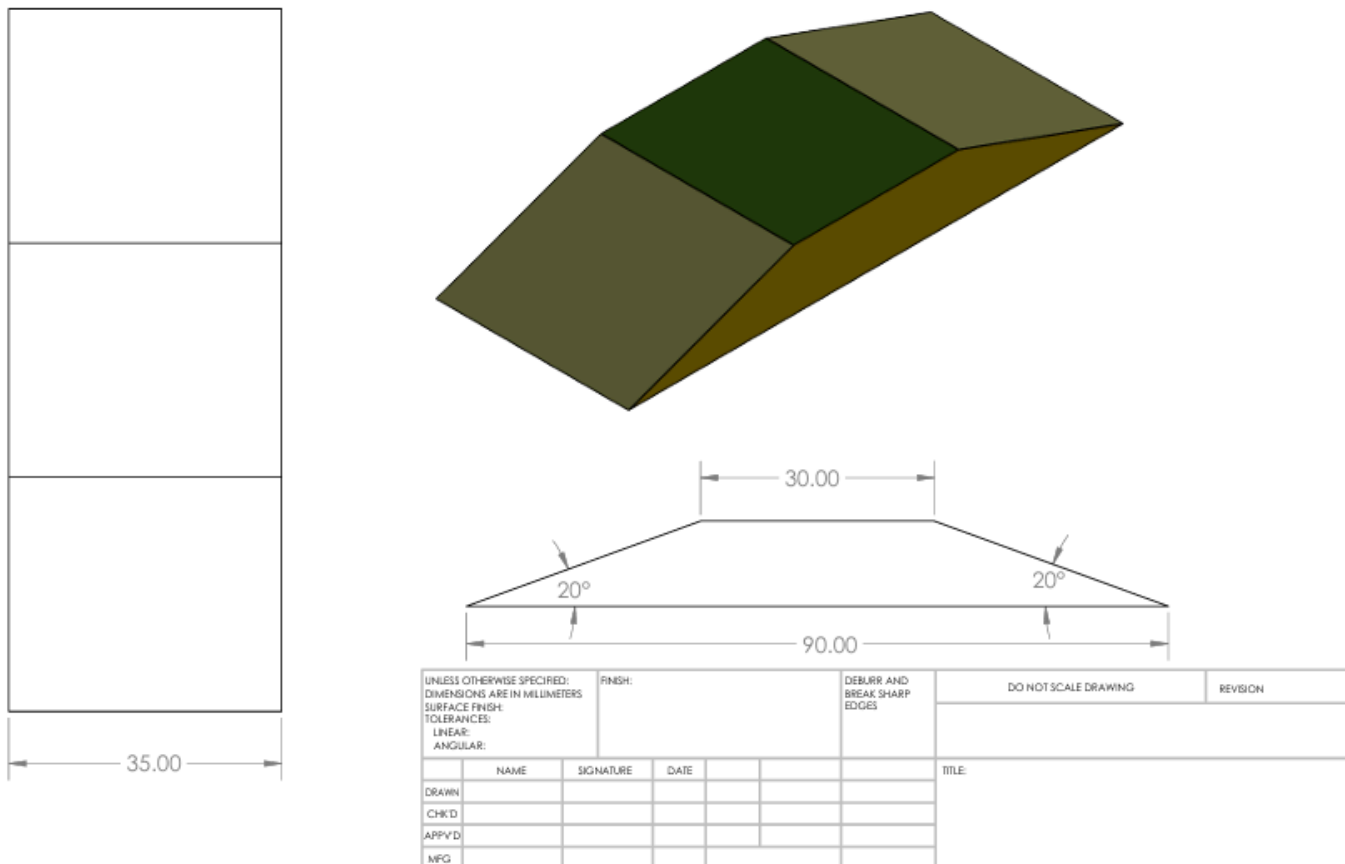
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Appendix 1 - For Task 1

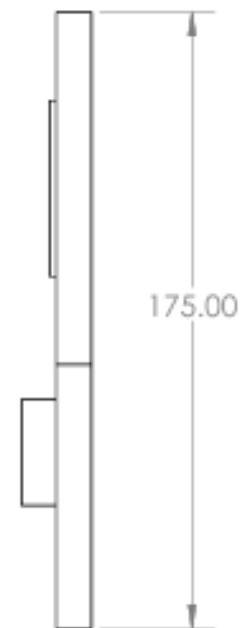
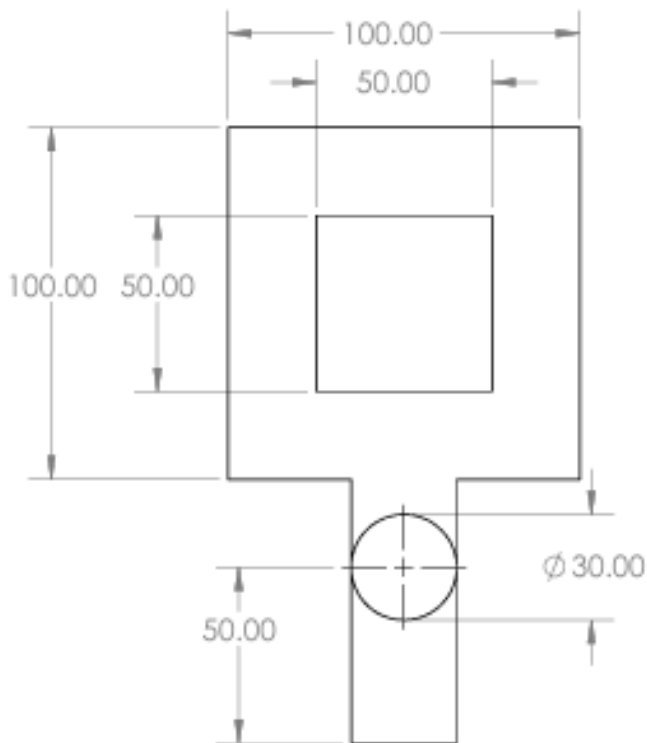
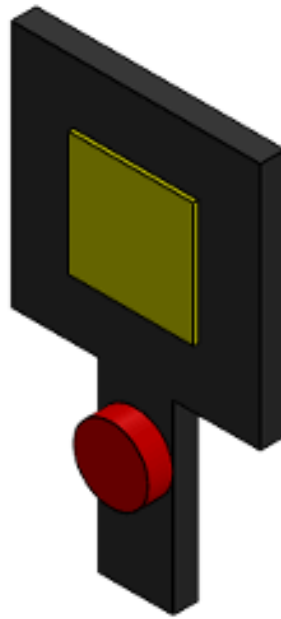


Appendix 2 - Ramp for Task 01



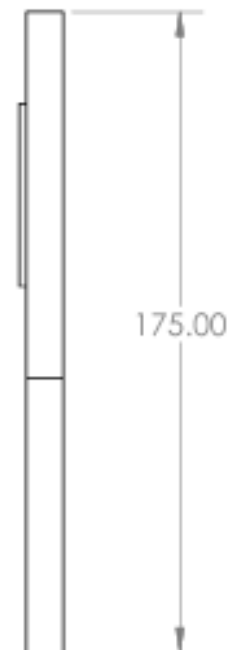
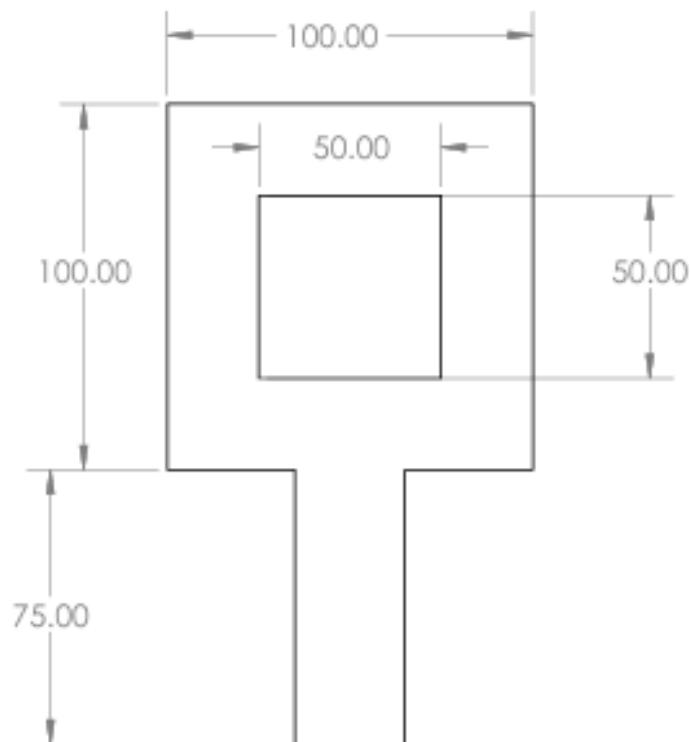
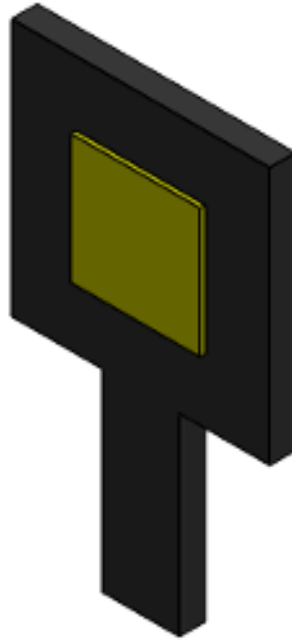
**For Appendix 2, all dimensions are in centimeters(cm)

Appendix 3 - Tower with button for Task 2



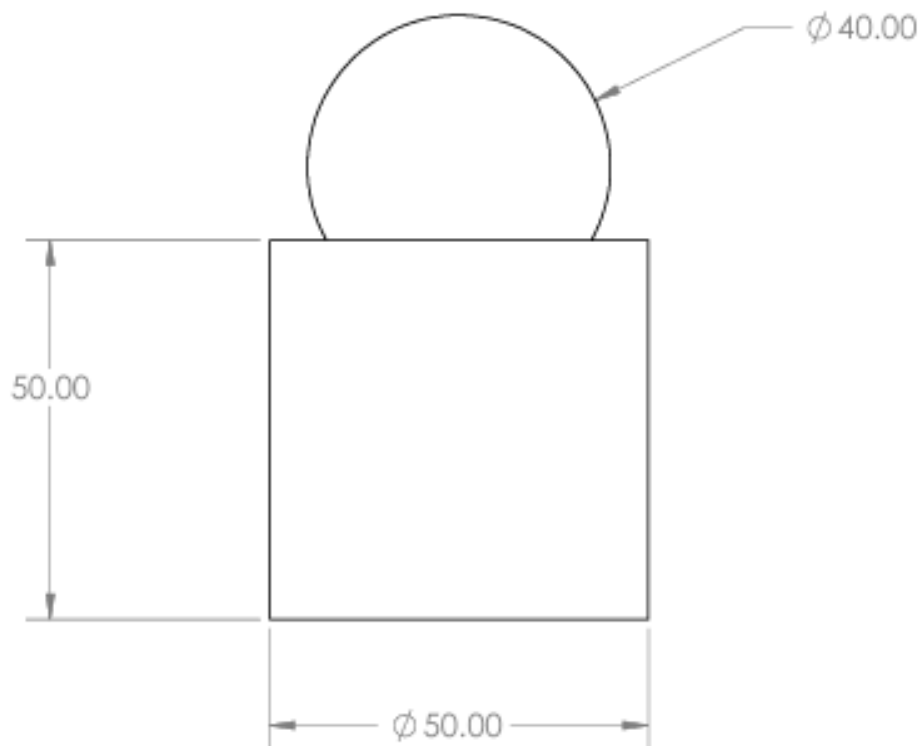
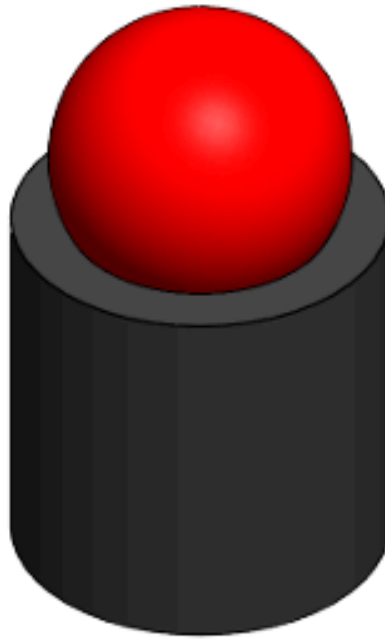
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:				FINISH:		DEBURE AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION	
	NAME	SIGNATURE	DATE					TITLE:			
DRAWN								DWG NO. Lamp with button ^{A4}			
CHECKED											
APPROVED											
MFG											
Q.A.											
				MATERIAL:							

Appendix 4 - Towers for Task 2

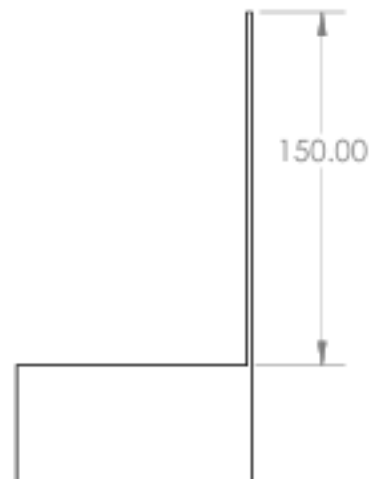
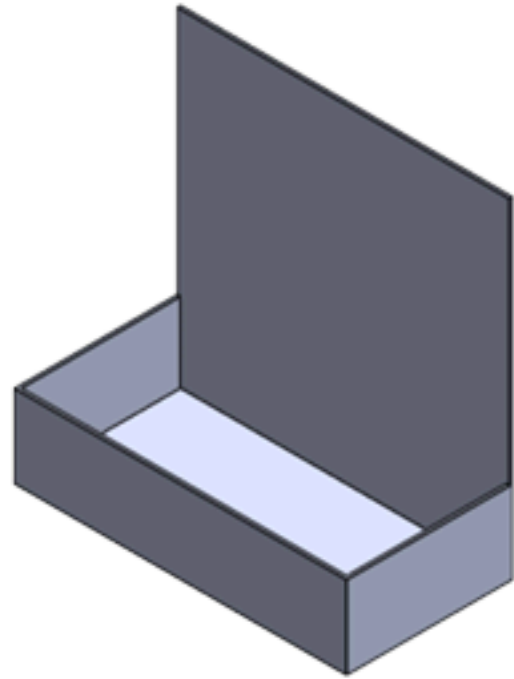
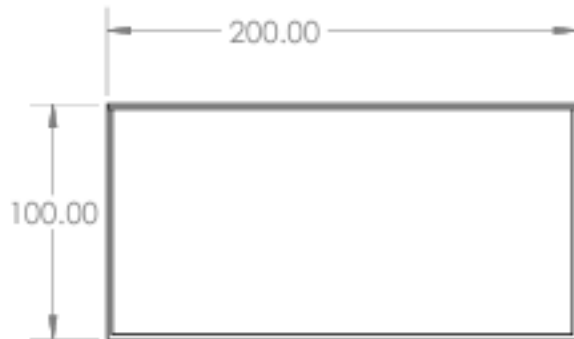


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: UNEAE: ANGULAR:		FINISH:				DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING		REVISION	
	NAME	SIGNATURE	DATE				TITLE:			
DRAWN										
CHKD										
APPROV										
MFG										
QA							MATERIAL:		DWG NO.	
									Lamp	
									A4	

Appendix 5 - Fragment and Fragment Holder

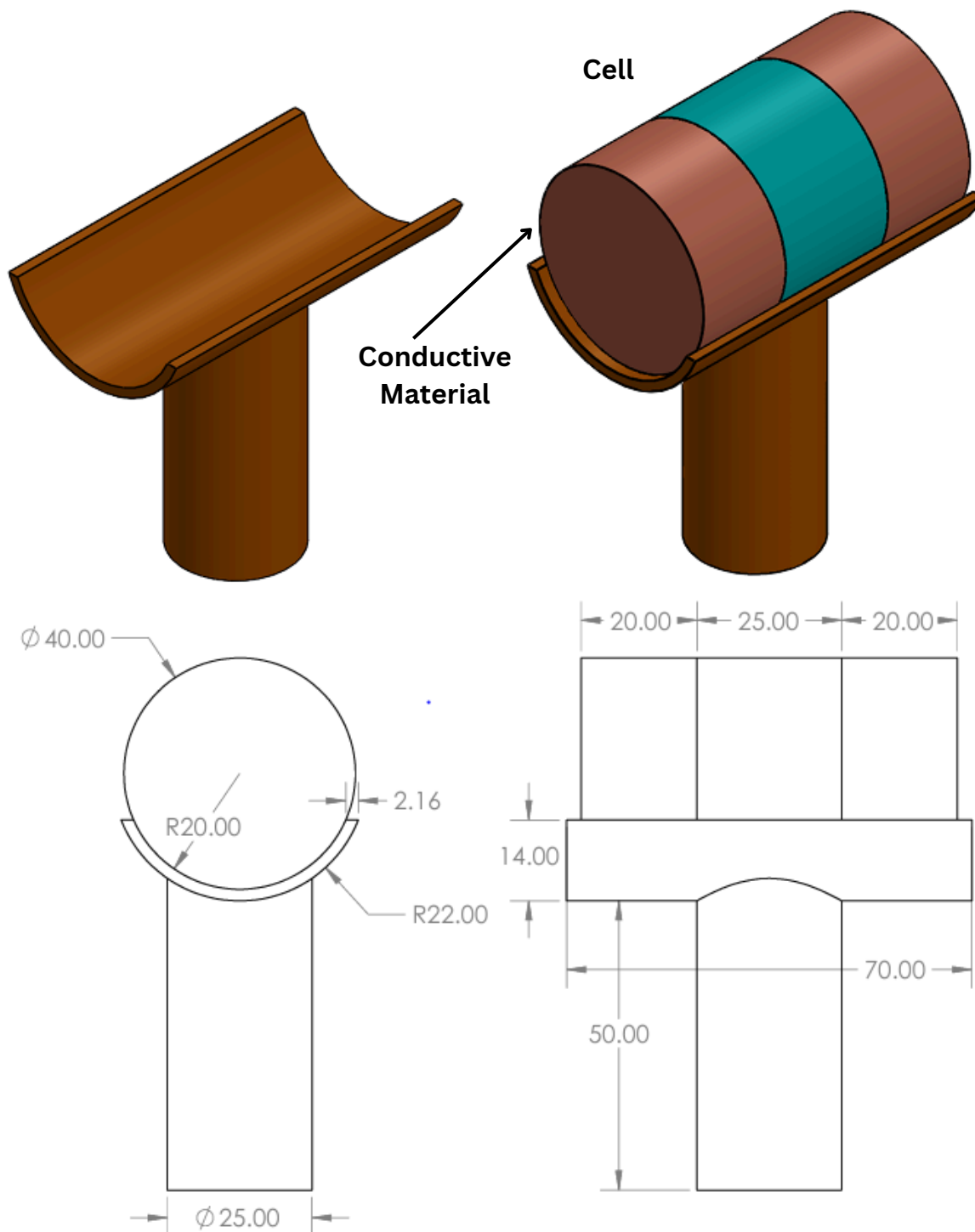


Appendix 6 - Storage container for Task 3



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:				DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE				TITLE:	
DRAWN								
CHECKED								
APPROVED								
MFG								

Appendix 7 - Cell and Cell holding towers



**cell holding tower and the charged cell placing tower will remain the same

**Both sides in the cells are covered with a conductive material

(Eg: Copper, Aluminium) and center part will be plastic and the color in the middle part will be blue